Chapter 20 Artificial Intelligence

After studying this chapter, the student should be able to:

- Define and give a brief history of artificial intelligence.
- Describe how knowledge is represented in an intelligent agent.
- Show how expert systems can be used when a human expert is not available.
- Show how an artificial agent can be used to simulate mundane tasks performed by human beings.
- Show how expert systems and mundane systems can use different search techniques to solve problems.
- Show how the learning process in humans can be simulated, to some extent, using neural networks that create the electronic version of a neuron called a *perceptron*.

20.1 INTRODUCTION

What is artificial intelligence?

Artificial intelligence is the study of programmed systems that can simulate, to some extent, human activities such as perceiving, thinking, learning and acting.

A brief history of artificial intelligence

- Although artificial intelligence as an independent field of study is relatively new, it has some roots in the past.
- We can say that it started 2,400 years ago when the Greek philosopher **Aristotle** invented the concept of logical reasoning.
- The effort to finalize the language of logic continued with **Leibniz** and **Newton**.
- **George Boole** developed Boolean algebra in the nineteenth century (Appendix E) that laid the foundation of computer circuits.
- However, the main idea of a thinking machine came from **Alan Turing**, who proposed the Turing test.
- The term “artificial intelligence” was first coined by John **McCarthy** in 1956.

The Turing test

- In 1950, Alan Turing proposed the Turing Test, which provides a definition of intelligence in a machine.
- The test simply compares the intelligent behavior of a human being with that of a computer.
- An interrogator asks a set of questions that are forwarded to both a computer and a human being.
- The interrogator receives two sets of responses, but does not know which set comes from the human and which set from the computer.
- After careful examination of the two sets, if the interrogator cannot definitely tell which set has come from the computer and which from the human, the computer has passed the Turing test for intelligent behavior.

Intelligent agents

- An **intelligent agent** is a system that perceives its environment, learns from it, and interacts with it intelligently. Intelligent agents can be divided into two broad categories:
- **Software agents**
  - A software agent is a set of programs that are designed to do particular tasks. For example, a software agent can check the contents of received e-mails and classifying them into different categories (junk, less important, important, very important, and so on). Another example of a software agent is a search engine used to search the World Wide Web and find sites that can provide information about a requested subject.
- **Physical agents**
  - A physical agent (robot) is a programmable system that can be used to perform a variety of tasks. Simple robots can be used in manufacturing to do routine jobs such as assembling, welding, or painting. Some organizations use mobile robots that do routine delivery jobs such as distributing mail or correspondence to different rooms. Mobile robots are used underwater to prospect for oil. A humanoid robot is an autonomous mobile robot that is supposed to behave like a human.

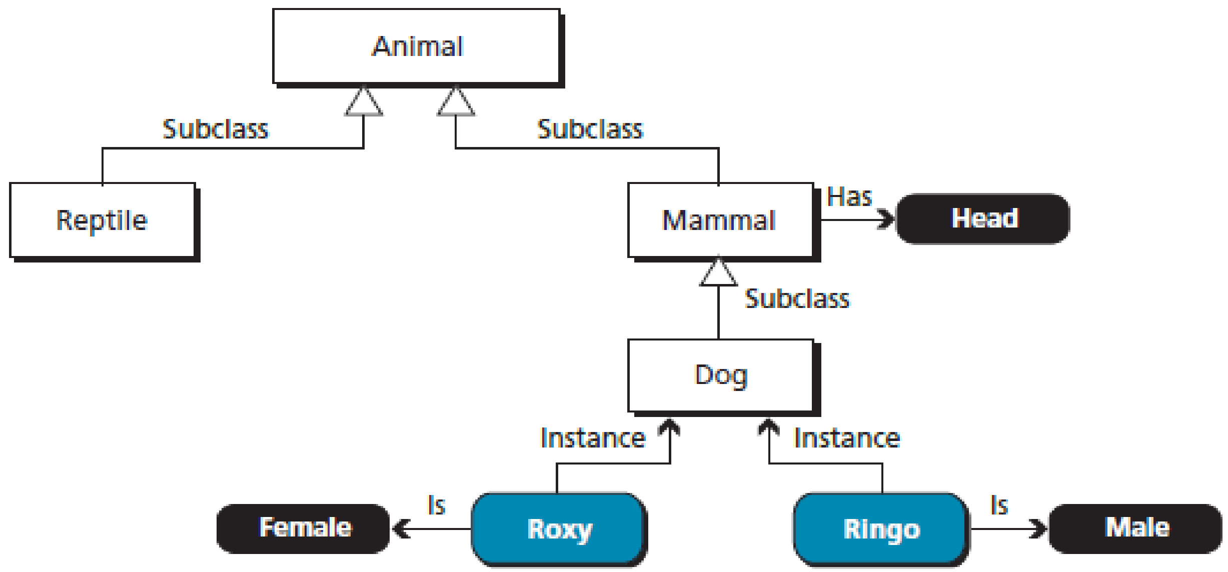
20.2 KNOWLEDGE REPRESENTATION

- If an artificial agent is supposed to solve some problems related to the real world, it needs to be able to represent knowledge somehow. Facts are represented as data structures that can be manipulated by programs stored inside the computer.
- In this section, we describe four common methods for representing knowledge:
  - **semantic networks**
  - **frames,**
  - **predicate logic**
  - **rule-based systems**

Semantic networks

- A semantic network uses directed graphs to represent knowledge.

Figure 20.1 *A simple semantic network*



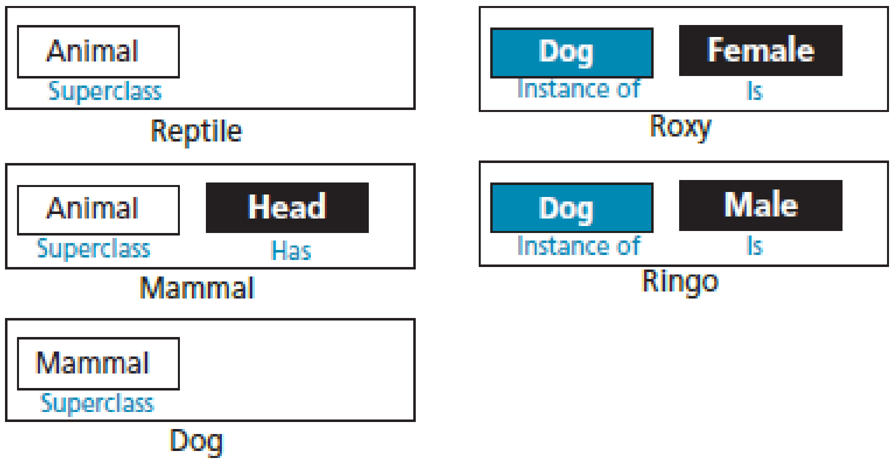
Concepts: A concept can be thought of as a set or a subset. For example, animal defines the set of all animals, horse defines the set of all horses and is a subset of the set animal.

Relations: In a semantic network, relations are shown by edges. An edge can define a subclass relation, an instance relation attribute, an object (color, size, ...), or a property of an object.

Frames

- Frames are closely related to semantic networks. In frames, data structures (records) are used to represent the same knowledge. One advantage of frames over semantic networks is that programs can handle frames more easily than semantic networks.

Figure 20.2 *A set of frames representing semantic network*



Objects: A node in a semantic network becomes an object in a set of frames, so an object can define a class, a subclass, or an instance of a class. In Figure 20.2, reptile, mammal, dog, Roxy, and Ringo are objects.

Slots: Edges in semantic networks are translated into slots—fields in the data structure. The name of the slot defines the type of the relationship and the value of the slot completes the relationship. In Figure 20.2, for example, animal is a slot in the reptile object.

Predicate logic

- Two quantifiers are common in predicate logic: ∀ “for all”, and ∃ “there exists”.

Deduction

Premiss 1:	All men are mortals.
Premiss 2:	Socrates is a man.
Conclusion:	Therefore, Socrates is mortal.

$$\forall x\ [\text{man}(x) \rightarrow \text{mortal}(x)],\ \text{man}(\text{Socrates}) \mid\!-\ \text{mortal}(\text{Socrates})$$

$$\text{man}(\text{Socrates}) \rightarrow \text{mortal}(\text{Socrates}),\ \text{man}(\text{Socrates}) \mid\!-\ \text{mortal}(\text{Socrates})$$

- Which is reduced to M1 → M2, M1 |− M2, in which M1 is man(Socrates) and M2 is mortal(Socrates).
- The result is an argument in propositional logic and can be easily validated.

Rule-based systems

- A rule-based system represents knowledge using a set of rules that can be used to deduce new facts from known facts. The rules express what is true if specific conditions are met.
- A rule-based database is a set of ***if… then…*** statements in the form

$$\text{If A then B} \quad \text{or} \quad A \rightarrow B$$

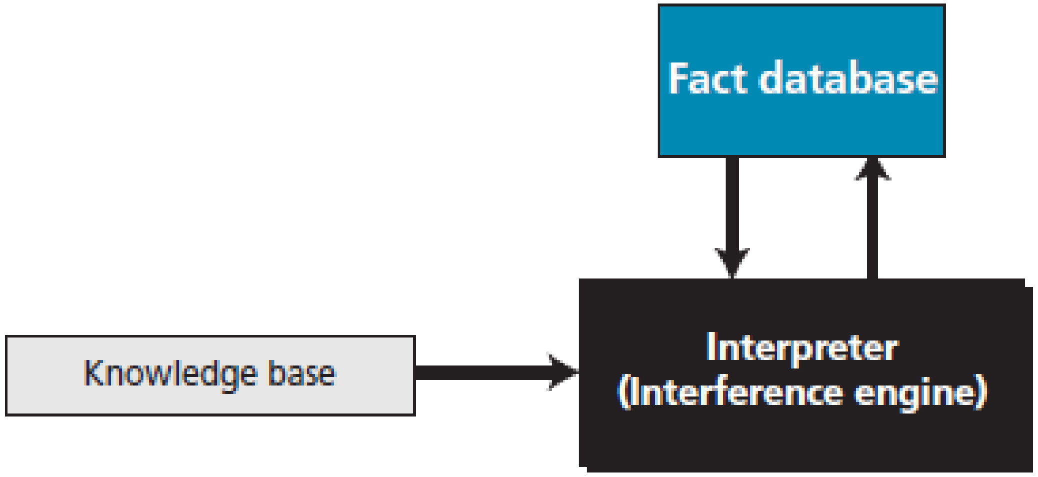
in which A is called the ***antecedent*** and B is called the ***consequent***.

- Note that in a rule-based system, each rule is handled independently without any connection to other rules.

Components

- A rule-based system is made up of three components: an interpreter (or inference engine), a knowledge base, and a fact database, as shown in Figure 20.4.

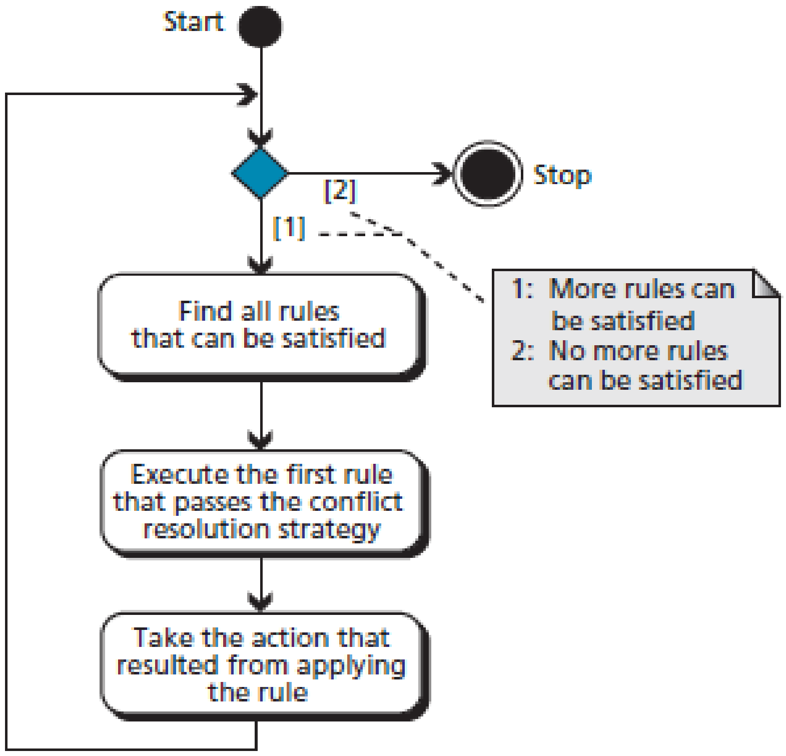
Figure 20.4 *The components of a rule-based system*



Forward chaining

- Forward chaining is the process in which an interpreter uses a set of rules and a set of facts to perform an action.

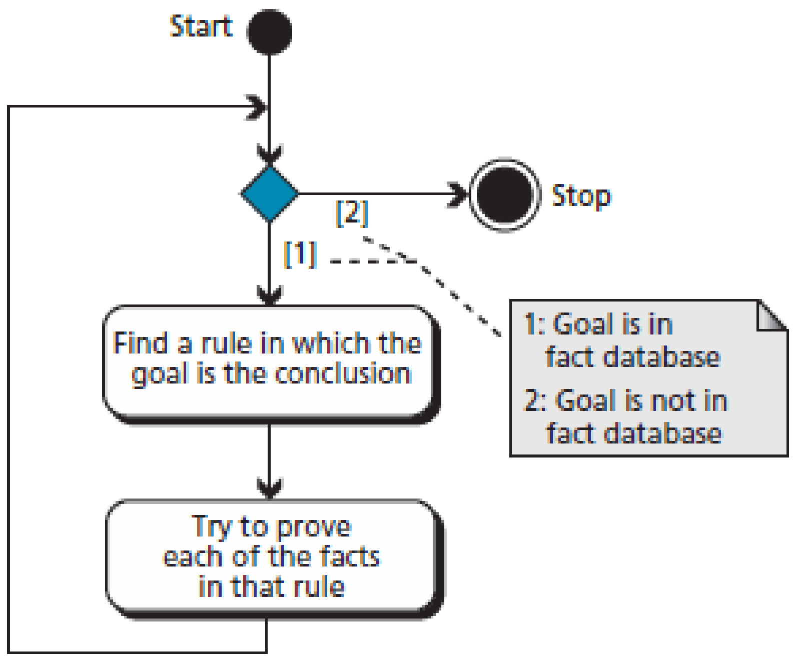
Figure 20.5 *Flow diagram for forward chaining*



Backward chaining

- Forward chaining is not very efficient if the system tries to prove a conclusion.
- In this cases, it may be more efficient if backward chaining is used.

Figure 20.6 *Flow diagram for backward chaining*



20.3 EXPERT SYSTEMS

- Expert systems use the knowledge representation languages discussed in the previous section to perform tasks that normally need human expertise.
- They can be used in situations in which that expertise is in short supply, expensive, or unavailable when required. For example, in medicine, an expert system can be used to narrow down a set of symptoms to a likely subset of causes, a task normally carried out by a doctor.

Extracting knowledge

- An expert system is built on predefined knowledge about its field of expertise. An expert system in medicine, for example, is built on the knowledge of a doctor specialized in the field for which the system is built: an expert system is supposed to do the same job as the human expert.
- The first step in building an expert system is therefore to extract the knowledge from a human expert. This extracted knowledge becomes the knowledge base we discussed in the previous section.

Extracting facts

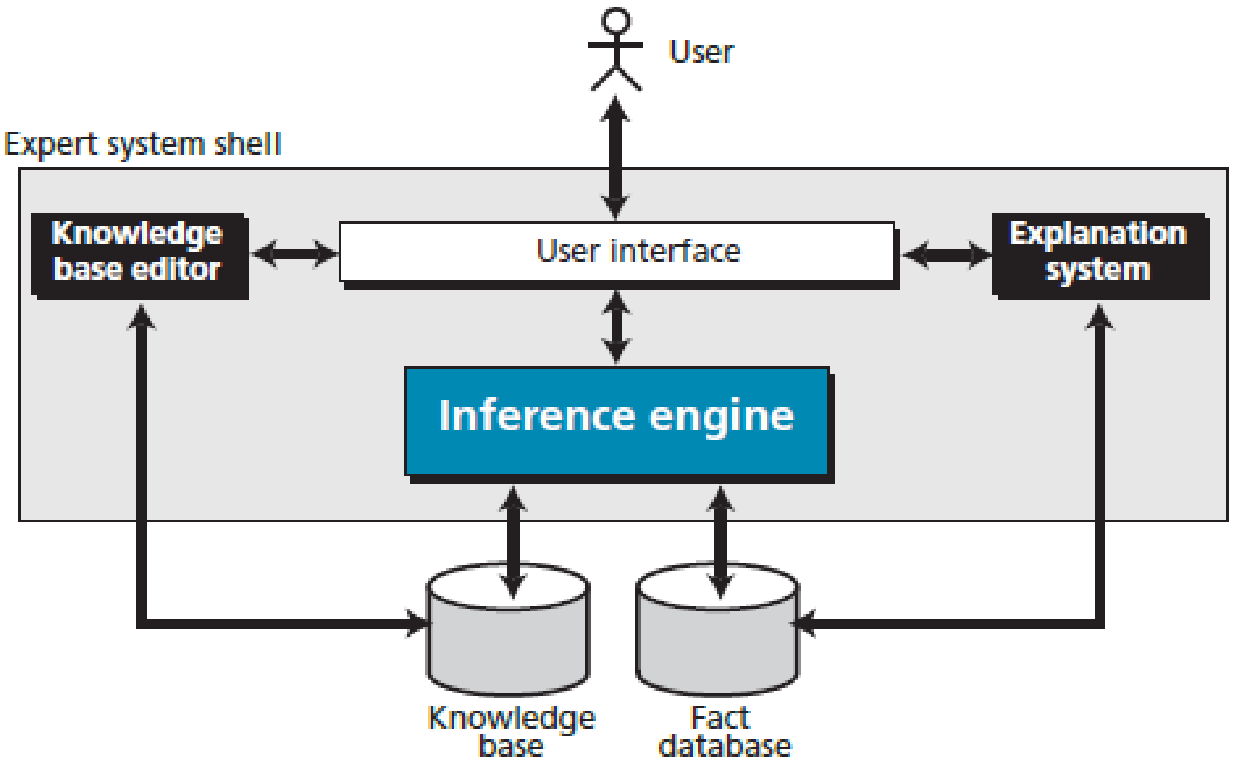
- To be able to infer new facts or perform actions, a fact database is needed in addition to the knowledge base for a knowledge representation language.
- The fact database in an expert system is case-based, in which facts collected or measured are entered into the system to be used by the inference engine.

Architecture

- Figure 20.7 shows the general idea behind the architecture of an expert system. As the figure shows, an expert system can have up to seven components: user, user interface, inference engine, knowledge base, fact database, explanation system, and knowledge base editor.
- The inference engine is the heart of an expert system: it communicates with the knowledge base, fact database, and the user interface. Four of the seven components of an expert system—user interface, inference engine, explanation system, and knowledge base editor—can be made once and used for many applications, as they are not dependent on the particular knowledge base or fact database.
- The figure shows these components in the shaded box, normally called an expert system shell.

Figure 20.7 shows the general idea behind the architecture of an expert system.

Figure 20.7 *The architecture of an expert system*



User: The user is the entity that uses the system to benefit from the expertise offered.

User Interface: The user interface allows the user to interact with the system. The user interface can accept natural language from the user and interpret it for the system. Most user interfaces also offer a user-friendly menu system.

Inference engine: The inference engine is the heart of the system that uses the knowledge base and the fact database to infer the action to be taken.

Knowledge Base: The knowledge base is a collection of knowledge based on interviews with experts in the relevant field of expertise.

Fact database: The fact database in an expert system is case-based. For each case, the user enters the available or measured data into the fact database to be used by the inference engine for that particular case.

Explanation system: The explanation system, which may not be included in all systems, is used to explain the rationale behind the decision made by the inference engine.

Knowledge base editor: The knowledge base editor, which may not be included in all systems, is used to update the knowledge base if new experience has been obtained from experts in the field.

20.4 PERCEPTION

- Another goal of AI is to create a machine that behaves like an ordinary human. One of the meanings of the word "perception" is understanding what is received through the senses—sight, hearing, touch, smell, taste.
- A human being sees a scene through the eyes, and the brain interprets it to extract the type of objects in the scene.
- A human being hears a set of voice signals through the ears, and the brain interprets it as a meaningful sentence, and so on.

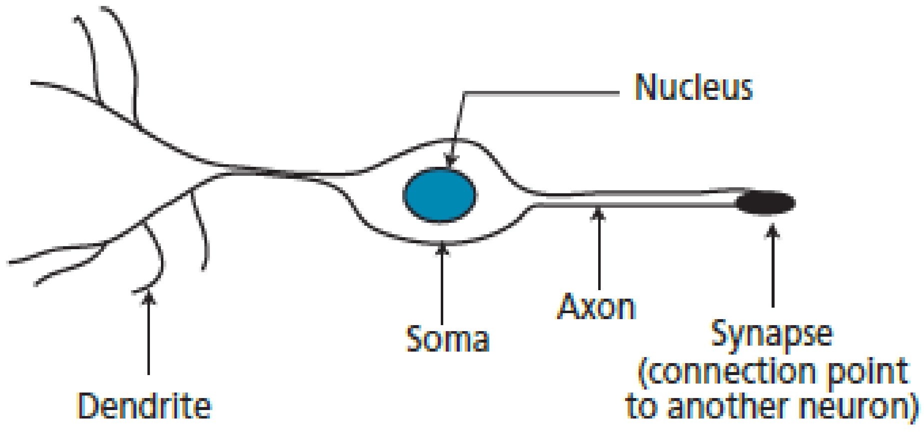
20.6 NEURAL NETWORKS

- If an intelligent agent is supposed to behave like a human being, it may need to learn.
- Learning is a complex biological phenomenon that is not even totally understood in humans.
- Enabling an artificial intelligence agent to learn is definitely not an easy task. However, several methods have been used in the past that create hope for the future.
- Most of the methods use inductive learning or learning by example. This means that a large set of problems and their solutions is given to the machine from which to learn.

Biological neurons

- The human brain has billions of processing units, called neurons.
- Each neuron, on average, is connected to several thousand other neurons.
- A neuron is made of three part: soma, axon, and dendrites, as shown in Figure 20.22.

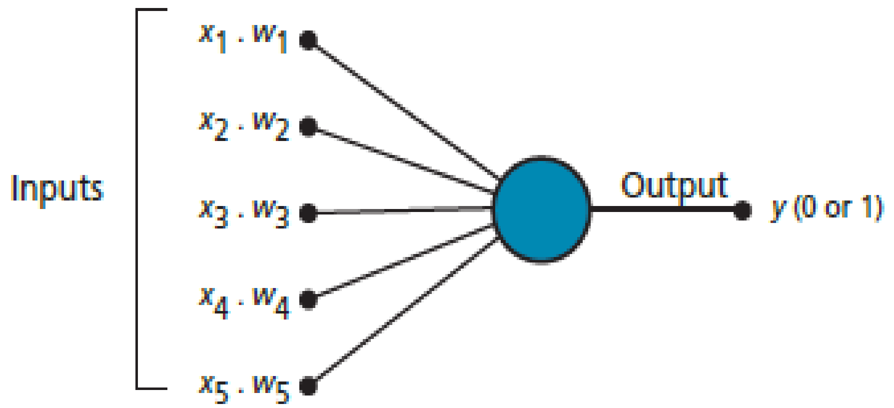
Figure 20.22 *A simplified diagram of a neuron*



Perceptrons

- A perceptron is an artificial neuron similar to a single biological neuron.
- It takes a set of weighted inputs, sums the inputs, and compares the result with a threshold value.
- If the result is above the threshold value, the perceptron fires, otherwise, it does not. When a perceptron fires, the output is 1: when it does not fire, the output is zero.

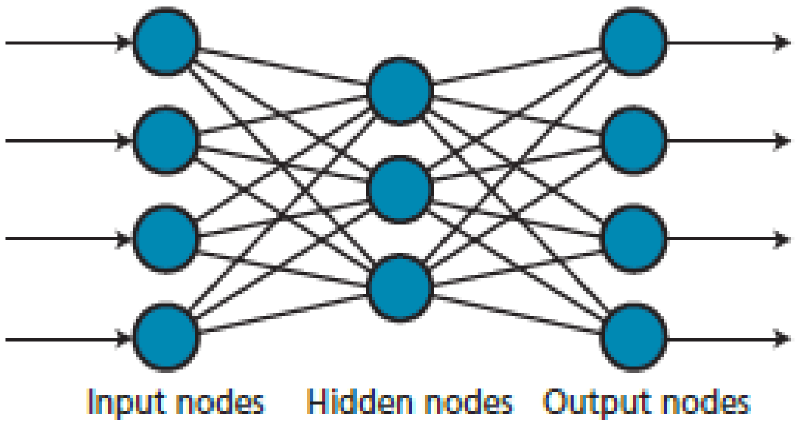
Figure 20.23 *A perceptron*



Multi-layer networks

- Several layers of perceptions can be combined to create multilayer neural networks. The output from each layer becomes the input to the next layer.
- The first layer is called the input layer, the middle layers are called the hidden layers, and the last layer is called the output layer.

Figure 20.24 *A multi-layer neural network*



Applications

- Neural networks can be used when enough pre-established inputs and outputs exist to train the network. Two areas in which neural networks have proved to be useful are **optical character recognition (OCR),** in which the intelligent agent is supposed to read any handwriting, and **credit assignment**, where different factors can be weighted to establish a credit rating, for example for a loan applicant.